This appendix provides the full proofs of Theorems 1–3 (Upper-bound Proofs, below), additional experimental results for $d \in \{3, 4, 5\}$ (Experimental details), and a summary of notation (Notation summary).

Upper-bound Proofs

This appendix establishes the sublinear α-regret guarantee for *THV-UCB* announced in the Regret Analysis section. We prove two complementary statements:

- a *gap-free* bound (Theorem 1), of order $\tilde{O}(\sqrt{nkT})$, valid on *every* instance regardless of how close the arms are to each other;
- a *gap-dependent* bound (Theorem 2), of order $\tilde{O}(nk^{2.5}/\Delta_{\min})$, which becomes polylogarithmic in T as soon as the instance is well separated.

Both proofs share the same first steps (Preliminaries through Lipschitz control of the optimism error), which reduce $\bar{R}_T$ to a sum of confidence radii, in (5). They diverge only in *how* this sum is controlled: the gap-free bound (see Section *A gap-free bound*) treats every pull anonymously via Cauchy–Schwarz over the whole horizon; the gap-dependent bound (See Section *A gap-dependent bound*) opens up the greedy construction stage by stage, tracks the actual comparisons $\Delta_t^{\mathrm{UCB}}(\cdot \mid S)$ performed by the algorithm, and applies the same Cauchy–Schwarz idea *locally*, to each stage's deviation rounds. We discuss what each proof technique does, and does not, use about *THV-UCB* in the discussion at the end of this appendix.

Preliminaries

Lemma 1 (Monotonicity and submodularity of hypervolume)**.** *For any fixed reference point* $\mathbf{r}$ *dominated by all achievable means, the set function* $S \mapsto \mathrm{HV}(S)$ *is monotone (non-decreasing) and submodular.*

Proof. Recall $\mathrm{HV}(S) = \lambda_d\big(\bigcup_{i\in S} \prod_{j=1}^{d} [r_j, \mu_{i,j}]\big)$.

Monotonicity. If $S \subseteq T$, the union of boxes only grows, so $\mathrm{HV}(S) \le \mathrm{HV}(T)$.

Submodularity. Let $A \subseteq B$ and $i \notin B$. Write $V_A = \bigcup_{j\in A} \prod_\ell [r_\ell, \mu_{j,\ell}]$ and $V_i = \prod_\ell [r_\ell, \mu_{i,\ell}]$. Then

$$\mathrm{HV}(A \cup \{i\}) - \mathrm{HV}(A) = \lambda_d(V_i \setminus V_A), \qquad \mathrm{HV}(B \cup \{i\}) - \mathrm{HV}(B) = \lambda_d(V_i \setminus V_B).$$

Since $V_A \subseteq V_B$, $V_i \setminus V_B \subseteq V_i \setminus V_A$, hence submodularity. □

Remark. *The same argument, applied verbatim with* $u_{i,j}(t)$ *in place of* $\mu_{i,j}$ *(the proof never uses anything about* $\mu_{i,j}$ *beyond it being a real number), shows that for every fixed* t, $S \mapsto \mathrm{HV}_t^{\mathrm{UCB}}(S)$ *is also monotone and submodular.*

Lemma 2 (Coordinate-wise concentration)**.** *Assume each coordinate is* η*-sub-Gaussian. Then with probability at least* $1 - \delta$*, for all* i, j, t*,*

$$\big|\widehat{\mu}_{i,j}(t) - \mu_{i,j}\big| \le \beta_i(t) = \sqrt{\frac{2\eta \log(ndt^2/\delta)}{\max\{1, N_i(t-1)\}}}.$$

Proof. Fix i, j and a pull count $s = N_i(t-1) \ge 1$. By the sub-Gaussian Hoeffding bound, $\mathbb{P}\big(|\widehat{\mu}_{i,j}(t) - \mu_{i,j}| > \sqrt{2\eta \log(x)/s}\big) \le 2/x$. Taking $x = ndt^2/\delta$ and a union bound over $i \in [n]$, $j \in [d]$, and $t \in [T]$ (using $\sum_{t\ge 1} t^{-2} \le 2$) gives total failure probability $O(\delta)$; absorbing the constant into δ yields the stated bound. □

We write $\mathcal{E}$ for the event on which Lemma 2 holds, so $\mathbb{P}(\mathcal{E}) \ge 1 - \delta$. After the initialization phase of Algorithm 1 (which lasts at most $t_0 = \lceil nm_0/k \rceil$ rounds), every arm has $N_i(t-1) \ge m_0 \ge 1$, so the $\max\{1, \cdot\}$ in $\beta_i(t)$ is never active for $t > t_0$; we drop it below for readability.

A shared greedy-approximation theorem

The following classical fact (Nemhauser, Wolsey & Fisher, 1978) is invoked twice in this appendix, for two different monotone submodular functions; we state it once to avoid duplicating the argument and, crucially, to keep visually distinct the two objects it produces.

Fact 1 (Greedy approximation for monotone submodular maximization)**.** *Let* $f : 2^{[n]} \to \mathbb{R}$ *be monotone and submodular, and let* $G = \{g_1, \ldots, g_k\}$ *be built by greedy maximization of marginal* f*-gain under the cardinality constraint* k *(i.e.* $g_\ell \in \arg\max_{i \notin G_{\ell-1}} f(G_{\ell-1} \cup \{i\}) - f(G_{\ell-1})$*,* $G_\ell = G_{\ell-1} \cup \{g_\ell\}$*,* $G_0 = \emptyset$*,* $G = G_k$*). Then*

$$f(G) \;\ge\; \Big(1 - \tfrac{1}{e}\Big) \max_{|S|=k} f(S) \;=\; \alpha \max_{|S|=k} f(S).$$

We will apply Fact 1 to two different functions:

- $f = \mathrm{HV}_t^{\mathrm{UCB}}$ (monotone submodular by the previous remark, with $G = S_t$ the set actually built by *THV-UCB* at round t — this yields Corollary 1 below, used in the reduction step;
- $f = \mathrm{HV}$ **directly on the true means** — a purely deterministic statement requiring no concentration event — with $G = G^\star$, a benchmark sequence introduced in Section *A gap-dependent bound* and **distinct from the regret's true optimum** $S^\star$.

Corollary 1 (Greedy approximation under optimism)**.** *For every round t, $\mathrm{HV}_t^{\mathrm{UCB}}(S_t) \geq \alpha\, \mathrm{HV}_t^{\mathrm{UCB}}(S_t^{\mathrm{opt}})$, where $S_t^{\mathrm{opt}} \in \arg\max_{|S|=k} \mathrm{HV}_t^{\mathrm{UCB}}(S)$.*

Optimism and reduction to optimism error

Recall $u_{i,j}(t) = \widehat{\mu}_{i,j}(t-1) + \beta_i(t)$ and

$$\mathrm{HV}_t^{\mathrm{UCB}}(S) = \lambda_d\Big(\bigcup_{i\in S}[r_1, u_{i,1}(t)] \times \cdots \times [r_d, u_{i,d}(t)]\Big).$$

On $\mathcal{E}$, $\mu_{i,j} \leq u_{i,j}(t)$ for all i, j, so by monotonicity $\mathrm{HV}(S) \leq \mathrm{HV}_t^{\mathrm{UCB}}(S)$ for every S. Let $S^\star \in \arg\max_{|S|=k} \mathrm{HV}(S)$, with $V^\star = \mathrm{HV}(S^\star)$ — this is the benchmark appearing in the regret definition (2), and the only role $S^\star$ plays in this appendix. Since S_t^{opt} maximizes $\mathrm{HV}_t^{\mathrm{UCB}}$ over *all* size-k sets, it dominates $S^\star$ under this score, so by Corollary 1,

$$\mathrm{HV}_t^{\mathrm{UCB}}(S_t) \;\geq\; \alpha\, \mathrm{HV}_t^{\mathrm{UCB}}(S_t^{\mathrm{opt}}) \;\geq\; \alpha\, \mathrm{HV}_t^{\mathrm{UCB}}(S^\star).$$

Therefore, on $\mathcal{E}$,

$$\bar{r}_t = \alpha\, V^\star - \mathrm{HV}(S_t) \;\leq\; \alpha\, \mathrm{HV}_t^{\mathrm{UCB}}(S^\star) - \mathrm{HV}(S_t) \;\leq\; \mathrm{HV}_t^{\mathrm{UCB}}(S_t) - \mathrm{HV}(S_t). \tag{3}$$

Summing over $t \in [T]$,

$$\bar{R}_T \;\leq\; \sum_{t=1}^{T}\big(\mathrm{HV}_t^{\mathrm{UCB}}(S_t) - \mathrm{HV}(S_t)\big) \qquad \text{on } \mathcal{E}. \tag{4}$$

Lipschitz control of the optimism error

Lemma 3 (Lipschitz bound for hypervolume under coordinate-wise shifts)**.** *There is a constant C_d (one may take $C_d \leq d$ when rewards lie in $[0,1]^d$ and $\mathbf{r} = \mathbf{0}$) such that, on $\mathcal{E}$, for every round t and* ***every finite set*** *$S \subseteq [n]$ (not necessarily S_t),*

$$\mathrm{HV}_t^{\mathrm{UCB}}(S) - \mathrm{HV}(S) \;\leq\; C_d \sum_{i\in S} \beta_i(t).$$

Proof. Write $S = \{i_1, \ldots, i_m\}$. Define the intermediate hypervolume A_ℓ where arms $i_1, \ldots, i_\ell$ use optimistic vectors $\mathbf{u}_{i_p}(t)$ and arms $i_{\ell+1}, \ldots, i_m$ use true means $\boldsymbol{\mu}_{i_p}$, so $\mathrm{HV}_t^{\mathrm{UCB}}(S) - \mathrm{HV}(S) = \sum_{\ell=1}^{m}(A_\ell - A_{\ell-1})$. Each term changes only arm i_ℓ from $\boldsymbol{\mu}_{i_\ell}$ to $\mathbf{u}_{i_\ell}(t) = \boldsymbol{\mu}_{i_\ell} + \boldsymbol{\delta}_{i_\ell}$ with $0 \leq \delta_{i_\ell,j} \leq \beta_{i_\ell}(t)$. The added hypervolume is contained in a union of d axis-aligned slabs of width at most $\beta_{i_\ell}(t)$ in one coordinate and at most 1 in the others, so $A_\ell - A_{\ell-1} \leq d\,\beta_{i_\ell}(t)$. Summing over ℓ gives the claim with $C_d = d$. Crucially, this argument never uses $|S| = k$ nor any property of how S was selected, hence it holds for an arbitrary finite S. □

Combining (4) with Lemma 3 applied to $S = S_t$,

$$\bar{R}_T \;\leq\; C_d \sum_{t=1}^{T}\sum_{i\in S_t} \beta_i(t) \qquad \text{on } \mathcal{E}. \tag{5}$$

The rest of the proof bounds the right-hand side of (5) in two ways.

A gap-free bound

Let $N_i(T) = \sum_{t=1}^{T} \mathbf{1}\{i \in S_t\}$. Reordering (5) by arm,

$$\sum_{t=1}^{T}\sum_{i\in S_t} \beta_i(t) = \sum_{i=1}^{n}\sum_{s=1}^{N_i(T)} \sqrt{\frac{2\eta\log(ndT^2/\delta)}{s}} \;\leq\; 2\sqrt{2\eta\log(ndT^2/\delta)}\sum_{i=1}^{n}\sqrt{N_i(T)},$$

using $\sum_{s=1}^{m} s^{-1/2} \leq 2\sqrt{m}$. Since exactly k arms are pulled per round, $\sum_i N_i(T) = kT$, and Cauchy–Schwarz gives $\sum_i \sqrt{N_i(T)} \leq \sqrt{nkT}$. Hence, on $\mathcal{E}$, $\bar{R}_T \leq O\big(C_d\sqrt{nkT\log T}\big)$.

Theorem 1 (Gap-free bound (detailed version))**.** *On $\mathcal{E}$, $\bar{R}_T = O\big(C_d\sqrt{nkT\log T}\big)$.*

Remark. *This argument never used that S_t is built by comparing marginal gains $\Delta_t^{\mathrm{UCB}}(i \mid S)$ across candidates: it only used $|S_t| = k$ and the parametric decay $\beta_i(t) = \Theta(1/\sqrt{N_i(t-1)})$. It is "gap-free" precisely because it is blind to how close arms are, hence to the selection mechanism itself.*

A gap-dependent bound

The key idea: the algorithm's own greedy construction at round t progressively learns a stage-by-stage benchmark sequence. At stage ℓ, it must identify the best arm conditionally on the previously identified greedy prefix.

The greedy optimal set Let $G^\star = \{i_1^\star, \dots, i_k^\star\}$ be a greedy construction of an optimal size-k set for the *true* hypervolume:

$$i_\ell^\star \in \arg\max_{i \notin G_{\ell-1}^\star} \Delta(i \mid G_{\ell-1}^\star), \qquad G_\ell^\star = G_{\ell-1}^\star \cup \{i_\ell^\star\}, \quad G_0^\star = \emptyset,$$

with $\Delta(i \mid S) = \mathrm{HV}(S \cup \{i\}) - \mathrm{HV}(S)$.

Assume the greedy maximizer is unique at every stage of $G^\star$'s construction, and define the stage-ℓ gaps relative to this benchmark:

$$\Delta_\ell(i) = \Delta(i_\ell^\star \mid G_{\ell-1}^\star) - \Delta(i \mid G_{\ell-1}^\star) > 0, \quad i \neq i_\ell^\star, \qquad \Delta_{\min} = \min_{\ell \in [k]} \min_{i \neq i_\ell^\star} \Delta_\ell(i).$$

Remark. *By submodularity (Lemma 1), the sequence of greedy* values $v_\ell := \Delta(i_\ell^\star \mid G_{\ell-1}^\star)$ *is non-increasing: for $i \notin G_\ell^\star$, $G_{\ell-1}^\star \subseteq G_\ell^\star$ gives $\Delta(i \mid G_\ell^\star) \leq \Delta(i \mid G_{\ell-1}^\star)$, so $v_{\ell+1} \leq v_\ell$. This only constrains the* best *marginal gain at each stage, not the runner-up: it does* not *imply that the gap sequence $\gamma_\ell := \min_{i \neq i_\ell^\star} \Delta_\ell(i)$ is monotonic in ℓ. We therefore use the safe, uniform bound $\Delta_{\min}$ rather than a stage-wise one.*

Deviation of optimistic marginal gains For $S \subseteq [n]$ and $i \notin S$, write $\Delta_t^{\mathrm{UCB}}(i \mid S) = \mathrm{HV}_t^{\mathrm{UCB}}(S \cup \{i\}) - \mathrm{HV}_t^{\mathrm{UCB}}(S)$.

Lemma 4 (Deviation of optimistic marginal gains)**.** *On $\mathcal{E}$, for every $S \subseteq [n]$ and $i \notin S$,*

$$\Delta_t^{\mathrm{UCB}}(i \mid S) \leq \Delta(i \mid S) + C_d \sum_{j \in S \cup \{i\}} \beta_j(t), \qquad \Delta_t^{\mathrm{UCB}}(i \mid S) \geq \Delta(i \mid S) - C_d \sum_{j \in S} \beta_j(t).$$

Proof. By Lemma 3 applied to $S \cup \{i\}$ and to S, $\mathrm{HV}_t^{\mathrm{UCB}}(S \cup \{i\}) \leq \mathrm{HV}(S \cup \{i\}) + C_d \sum_{j \in S \cup \{i\}} \beta_j(t)$, while optimism gives $\mathrm{HV}_t^{\mathrm{UCB}}(S) \geq \mathrm{HV}(S)$; subtracting gives the upper bound. Symmetrically, $\mathrm{HV}_t^{\mathrm{UCB}}(S \cup \{i\}) \geq \mathrm{HV}(S \cup \{i\})$ and $\mathrm{HV}_t^{\mathrm{UCB}}(S) \leq \mathrm{HV}(S) + C_d \sum_{j \in S} \beta_j(t)$ give the lower bound. □

Per-round deviation stage Write the algorithm's own greedy construction at round t as a chain $\hat{S}_{t,0} = \emptyset \subset \hat{S}_{t,1} \subset \cdots \subset \hat{S}_{t,k} = S_t$, where $\hat{S}_{t,\ell} = \hat{S}_{t,\ell-1} \cup \{\hat{\imath}_{t,\ell}\}$ and $\hat{\imath}_{t,\ell} \in \arg\max_{i \in A_t \setminus \hat{S}_{t,\ell-1}} \Delta_t^{\mathrm{UCB}}(i \mid \hat{S}_{t,\ell-1})$.[3] Define the *first deviation stage*, comparing the algorithm's trajectory to the benchmark $G^\star$:

$$\ell^\dagger(t) = \min\{\ell \in [k] : \hat{S}_{t,\ell} \neq G_\ell^\star\},$$

with $\ell^\dagger(t) = +\infty$ if $\hat{S}_{t,\ell} = G_\ell^\star$ for every $\ell \in [k]$ (i.e. $S_t = G^\star$ exactly). This partitions $[T]$ into

$$\mathrm{Match} = \{t : \ell^\dagger(t) = \infty\}, \qquad \mathrm{Dev}_\ell = \{t : \ell^\dagger(t) = \ell\} \ (\ell \in [k]), \qquad [T] = \mathrm{Match} \sqcup \bigsqcup_{\ell=1}^{k} \mathrm{Dev}_\ell.$$

By construction, $t \in \mathrm{Dev}_\ell$ means $\hat{S}_{t,\ell-1} = G_{\ell-1}^\star$ — this is built into the partition, no induction over earlier rounds is needed.

Since $\bar{R}_T = \sum_{t=1}^T \bar{r}_t$ exactly by (2), this partition refines the same per-round sum bounded pointwise in (3): rather than controlling every term $\bar{r}_t$ uniformly via Lemma 3 and a single global Cauchy–Schwarz step (as in the gap-free bound above), we now split

$$\bar{R}_T \;=\; \sum_{t \in \mathrm{Match}} \bar{r}_t \;+\; \sum_{\ell=1}^{k} \sum_{t \in \mathrm{Dev}_\ell} \bar{r}_t, \tag{6}$$

and bound the two kinds of terms separately: we show below that the first sum is non-positive; the remainder of this subsection bounds the second.

Regret on matched rounds is non-positive We bound the first term of (6). If $t \in \mathrm{Match}$, then $S_t = G^\star$. By Fact 1 with $f = \mathrm{HV}$, $G = G^\star$,

$$\bar{r}_t = \alpha\, V^\star - \mathrm{HV}(S_t) = \alpha\, V^\star - \mathrm{HV}(G^\star) \;\leq\; \alpha\, V^\star - \alpha\, V^\star = 0.$$

[3] We assume $G^\star \subseteq A_t$, i.e. that the safe-pruning step never removes an arm that the true-objective greedy benchmark would have selected; this is the intended behavior of the pruning rule but is not separately proved here.

Witness-counting bound on deviation rounds Fix ℓ and $t \in \mathrm{Dev}_\ell$, and write $i := \hat{\imath}_{t,\ell} \neq i^\star_\ell$. Since i was chosen over $i^\star_\ell$ at stage ℓ of round t, $\Delta^{\mathrm{UCB}}_t(i \mid G^\star_{\ell-1}) \geq \Delta^{\mathrm{UCB}}_t(i^\star_\ell \mid G^\star_{\ell-1})$. Applying Lemma 4 at $S = G^\star_{\ell-1}$,

$$\Delta(i \mid G^\star_{\ell-1}) + C_d \sum_{j \in G^\star_{\ell-1} \cup \{i\}} \beta_j(t) \;\geq\; \Delta(i^\star_\ell \mid G^\star_{\ell-1}) - C_d \sum_{j \in G^\star_{\ell-1}} \beta_j(t),$$

so, using $|G^\star_{\ell-1}| = \ell - 1$,

$$\Delta_\ell(i) \;\leq\; C_d\Big(2 \sum_{j \in G^\star_{\ell-1}} \beta_j(t) + \beta_i(t)\Big) \;\leq\; C_d(2\ell - 1) \max_{j \in G^\star_{\ell-1} \cup \{i\}} \beta_j(t).$$

Hence there exists a *witness* $w(t) \in G^\star_{\ell-1} \cup \{i\}$ (at most ℓ candidates) with

$$\beta_{w(t)}(t) \;\geq\; \frac{\Delta_\ell(i)}{C_d(2\ell-1)} \;\geq\; \frac{\Delta_{\min}}{C_d(2\ell-1)},$$

$$\text{so} \qquad N_{w(t)}(t-1) \;\leq\; m_\ell := \frac{2\eta C_d^2 (2\ell-1)^2 \log(ndT^2/\delta)}{\Delta^2_{\min}} = O\!\left(\frac{\ell^2 \log T}{\Delta^2_{\min}}\right).$$

Since $G^\star_{\ell-1} \cup \{i\} = \hat{S}_{t,\ell} \subseteq S_t$, the witness $w(t)$ is pulled at round t. Group Dev_ℓ by witness identity: for fixed $w \in [n]$, every $t \in \mathrm{Dev}_\ell$ with $w(t) = w$ increments N_w by exactly 1 while $N_w(t-1) \leq m_\ell$; there are at most $m_\ell + 1$ such rounds. Summing over $w \in [n]$,

$$|\mathrm{Dev}_\ell| \;\leq\; n\,(m_\ell + 1) \;=\; O\!\left(\frac{n\ell^2 \log T}{\Delta^2_{\min}}\right). \tag{7}$$

Bounding total regret on deviation rounds We now bound the second term of (6), $\sum_{\ell=1}^{k} \sum_{t \in \mathrm{Dev}_\ell} \bar{r}_t$. A naive approach would bound $\bar{r}_t \leq 1$ on every round of Dev_ℓ and multiply by the count $|\mathrm{Dev}_\ell| = O(n\ell^2 \log T/\Delta^2_{\min})$ from (7), giving a regret scaling as $1/\Delta^2_{\min}$. We avoid this by bounding the *total* regret over Dev_ℓ directly, via the same Cauchy–Schwarz idea as in the *gap-free bound* Section above — applied *locally* to Dev_ℓ rather than to the full horizon. Since that argument scales as the *square root* of the number of rounds involved, it converts the $1/\Delta^2_{\min}$ scale of the count into a $1/\Delta_{\min}$ contribution to regret.

On $\mathcal{E}$, the pointwise bound (3) together with Lemma 3 gives, for every round t,

$$\bar{r}_t \;\leq\; \mathrm{HV}^{\mathrm{UCB}}_t(S_t) - \mathrm{HV}(S_t) \;\leq\; C_d \sum_{i \in S_t} \beta_i(t).$$

Fix $\ell \in [k]$, and for $i \in [n]$ let $D_i = \{t \in \mathrm{Dev}_\ell : i \in S_t\}$ be the (possibly empty) set of rounds, among those at which the trajectory first departs from $G^\star$ at stage ℓ, where arm i happens to be pulled. The pull count $N_i(t-1)$ increases by exactly 1 at each pull of i, so as t ranges over D_i (a set of pulls of i), the values $N_i(t-1)$ are pairwise distinct; hence their j-th smallest value is at least $j - 1$. Since $\beta_i(t) \propto N_i(t-1)^{-1/2}$ is decreasing, this bounds the sum by the worst case where D_i consists of arm i's *earliest* $|D_i|$ pulls:

$$\sum_{t \in D_i} \beta_i(t) \;\leq\; \sum_{s=1}^{|D_i|} \sqrt{\frac{2\eta \log(ndT^2/\delta)}{s}} \;\leq\; 2\sqrt{2\eta \log(ndT^2/\delta)}\,\sqrt{|D_i|} \;=\; O\big(\sqrt{|D_i| \log T}\big),$$

using $\sum_{s=1}^{m} s^{-1/2} \leq 2\sqrt{m}$ and $\log(ndT^2/\delta) = O(\log T)$ — the same argument as in the *gap-free bound* Section, restricted to $D_i \subseteq \{\text{pulls of } i\}$ instead of all of i's pulls over $[T]$.

Hence

$$\sum_{t \in \mathrm{Dev}_\ell} \sum_{i \in S_t} \beta_i(t) = \sum_{i=1}^{n} \sum_{t \in D_i} \beta_i(t) = O\Big(\sqrt{\log T} \sum_{i=1}^{n} \sqrt{|D_i|}\Big).$$

Since $|S_t| = k$ for every t, we have $\sum_{i=1}^{n} |D_i| = \sum_{t \in \mathrm{Dev}_\ell} \sum_{i=1}^{n} \mathbf{1}\{i \in S_t\} = \sum_{t \in \mathrm{Dev}_\ell} |S_t| = k|\mathrm{Dev}_\ell|$. Next, by Cauchy–Schwarz (with $a_i = 1, b_i = \sqrt{|D_i|}$),

$$\sum_{i=1}^{n} \sqrt{|D_i|} \;\leq\; \sqrt{n}\,\sqrt{\sum_{i=1}^{n} |D_i|} \;=\; \sqrt{nk|\mathrm{Dev}_\ell|}.$$

Therefore

$$\sum_{t \in \mathrm{Dev}_\ell} \bar{r}_t \;\leq\; C_d \sum_{t \in \mathrm{Dev}_\ell} \sum_{i \in S_t} \beta_i(t) \;=\; O\big(\sqrt{nk|\mathrm{Dev}_\ell| \log T}\big) \;=\; O\!\left(\frac{n\ell\sqrt{k}\,\log T}{\Delta_{\min}}\right),$$

using (7) in the last step. Summing over $\ell \in [k]$ (using $\sum_{\ell=1}^{k} \ell = O(k^2)$) and combining with (6) and the non-positive first term,

$$\bar{R}_T \;\le\; \sum_{\ell=1}^{k} O\!\left(\frac{n\ell\sqrt{k}\log T}{\Delta_{\min}}\right) \;=\; O\!\left(\frac{nk^{2.5}\log T}{\Delta_{\min}}\right) \qquad \text{on } \mathcal{E}.$$

Theorem 2 (Gap-dependent bound (detailed version))**.** *On $\mathcal{E}$, $\bar{R}_T = O\big(nk^{2.5}\log T/\Delta_{\min}\big)$.*

Combining both bounds

Since rewards lie in $[0,1]^d$, $\bar{r}_t \in [0,1]$ and $\bar{R}_T \le T$ always (on or off $\mathcal{E}$). Hence

$$\mathbb{E}[\bar{R}_T] = \mathbb{E}[\bar{R}_T \mathbf{1}_{\mathcal{E}}] + \mathbb{E}[\bar{R}_T \mathbf{1}_{\mathcal{E}^c}] \le \min\Big\{O\big(C_d\sqrt{nkT\log T}\big),\, O\big(nk^{2.5}\log T/\Delta_{\min}\big)\Big\} + T\,\mathbb{P}(\mathcal{E}^c).$$

Choosing $\delta = 1/T$ gives $T\,\mathbb{P}(\mathcal{E}^c) \le 1$, so:

Theorem 3 (Regret of *THV-UCB* (detailed version))**.** *With $\delta = 1/T$,*

$$\mathbb{E}[\bar{R}_T] \;\le\; \min\Big\{O\big(\sqrt{nkT\log T}\big),\, O\big(nk^{2.5}\log T/\Delta_{\min}\big)\Big\} + O(1).$$

In particular $\mathbb{E}[\bar{R}_T]/T \to 0$: THV-UCB achieves sublinear α-approximation regret.

Discussion: what does each bound use from the algorithm?

The two guarantees of Theorem 3 are not in competition; they cover complementary regimes, exactly as the minimax and gap-dependent bounds do for classical K-armed UCB. The gap-free bound (Theorem 1) is *algorithm-agnostic*: its proof only uses that k arms are pulled per round and that confidence widths shrink at rate $1/\sqrt{N_i(t-1)}$; it never degrades, even as $\Delta_{\min} \to 0$, the regime where the gap-dependent bound diverges. Conversely, the gap-dependent bound (Theorem 2) is *mechanism-aware*: it tracks, round by round, the exact comparisons $\Delta_t^{\mathrm{UCB}}(\cdot \mid G^\star_{\ell-1})$ that the greedy step of *THV-UCB* performs against the benchmark trajectory $G^\star$ — note that $G^\star$, unlike $S^\star$, never appears in the regret's definition; it is purely an internal proof device for tracking the algorithm's stage-wise progress. Its proof combines two ingredients: a witness-counting argument bounding how many rounds can deviate from $G^\star$ at each stage (see (7)), and a Cauchy–Schwarz argument bounding the *total* regret accrued on those rounds.

Experimental details

Optimal values (Grid search)

	$d=2$				$d=3$				$d=4$				$d=5$			
Algorithm	*Clust.*	*Conc.*	*Conv.*	*Lin.*	*Clust.*	*Conc.*	*Conv.*	*Lin.*	*Clust.*	*Conc.*	*Conv.*	*Lin.*	*Clust.*	*Conc.*	*Conv.*	*Lin.*
THV-UCB (ours)	0.01	0.01	0.01	0.01	0.01	0.01	0.01	0.01	0.01	0.01	0.01	0.01	0.01	0.01	0.01	0.01
ParetoUCB$^+$	1.0	1.0	1.0	1.0	0.01	0.1	0.1	0.3	0.01	0.1	0.3	0.3	0.01	0.01	0.01	0.1
ParetoUCB-Div	0.3	0.1	1.0	0.01	1.0	0.3	0.3	0.3	0.01	0.01	0.01	0.3	0.01	0.01	1.0	0.1
ParetoUCB-Crowd	1.0	1.0	1.0	0.3	1.0	0.01	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0
ChebyshevUCB	0.01	0.01	1.0	0.01	0.01	0.01	1.0	0.01	0.01	0.01	0.01	0.01	0.01	0.01	0.01	0.01
ChebyshevUCB$^+$	0.3	1.0	1.0	0.3	0.1	0.3	1.0	0.01	0.01	0.01	0.3	0.01	0.01	0.01	0.1	0.3
HVScalarUCB	0.01	0.01	1.0	0.01	0.01	0.01	0.01	0.01	0.01	0.01	0.01	0.3	0.01	0.01	0.01	0.01
HVScalarUCB$^+$	0.01	0.01	0.3	0.01	0.01	0.01	0.01	0.01	0.1	0.01	0.01	0.01	0.01	0.01	0.01	0.01
ScalarUCB	0.3	1.0	1.0	1.0	0.01	0.1	0.1	0.1	0.01	0.1	0.3	0.1	0.01	0.01	0.01	0.01
ScalarUCB-RandW	1.0	1.0	1.0	1.0	1.0	0.01	1.0	1.0	0.01	0.01	0.01	1.0	0.01	0.01	0.01	0.01

Table 2: Optimal confidence parameter $\eta^\star$ per algorithm and front geometry, selected by grid search over $\{0.01, 0.1, 0.3, 1.0\}$ (maximizing $\mathrm{HV}_{\mathrm{last100}}$). All detailed results are available in https://ngutowski.fr/gridsearch/gridsearch.html

Additional results for $d = 3$ and $k = 4$

		Clusters		**Concave**		**Convex**		**Linear**	
Algorithm	**Fid.**	$\mathrm{HV}_{\mathrm{last100}}$	Final Regret	$\mathrm{HV}_{\mathrm{last100}}$	Final Regret	$\mathrm{HV}_{\mathrm{last100}}$	Final Regret	$\mathrm{HV}_{\mathrm{last100}}$	Final Regret
THV-UCB (ours)	—	**0.2955**±0.001	**2.8**<0.001	**0.2894**±0.002	**2.2**<0.001	**0.5895**±0.003	**2.5**<0.001	**0.1592**±0.001	1.9±0.1
ParetoUCB (Drugan and Nowe 2013)	•	0.1577±0.034	141.1±80.8	0.1757±0.034	104.9±78.5	0.4648±0.043	45.4±55.1	0.0610±0.020	133.9±55.1
ParetoUCB$^+$ (Drugan and Nowe 2013)	○	0.2844±0.008	**2.8**<0.001	0.2755±0.005	3.2±1.3	0.5476±0.011	24.3±2.0	0.0934±0.018	59.4±40.1
ParetoUCB-Div (Drugan and Nowe 2013)	○	0.2083±0.025	71.9±52.2	0.1988±0.018	33.2±36.5	0.5161±0.012	6.1±1.3	0.0786±0.014	93.8±46.6
ParetoUCB-Crowd (Drugan and Nowe 2013; Deb et al. 2002)	○	0.1694±0.021	62.1±39.0	0.1290±0.008	135.7±41.8	0.4755±0.038	4.6±3.4	0.0251±0.006	184.3±28.6
ParetoTS (Yahyaa and Manderick 2015)	•	0.1708±0.031	93.5±63.4	0.1883±0.030	60.4±55.2	0.4836±0.038	15.4±20.9	0.0743±0.020	91.7±50.4
ParetoTS$^+$ (Yahyaa and Manderick 2015)	○	0.2690±0.011	12.4±10.4	0.2678±0.008	7.4±6.8	0.5465±0.009	3.0±0.8	0.0868±0.019	73.4±45.1
ChebyshevUCB (Mandow, Martín-Albo, and Perez-de-la Cruza 2023)	•	0.1296±0.055	246.0±113.3	0.1968±0.026	50.2±26.6	0.4322±0.035	23.0±30.6	0.0980±0.020	62.2±32.8
ChebyshevUCB$^+$ (Mandow, Martín-Albo, and Perez-de-la Cruza 2023)	○	0.2639±0.008	3.7±0.9	0.2727±0.004	4.4±1.3	0.5437±0.008	9.0±4.0	0.1401±0.001	**1.9**±0.1
HVScalarUCB (Zhang and Golovin 2020)	•	0.2046±0.036	76.7±62.1	0.2294±0.027	33.2±37.7	0.4526±0.044	41.4±55.7	0.1252±0.017	27.9±29.0
HVScalarUCB$^+$ (Zhang 2024)	•	0.2500±0.018	9.6±10.8	0.2448±0.018	9.1±10.7	0.4956±0.032	7.7±8.8	0.1357±0.010	6.8±7.3
ScalarUCB (Auer, Cesa-Bianchi, and Fischer 2002)	○	0.2807±0.010	**2.8**<0.001	0.2757±0.005	3.9±2.1	0.5472±0.007	**2.6**±0.1	0.0891±0.019	68.5±43.7
ScalarUCB-RandW (Paria, Kandasamy, and Póczos 2020)	○	0.1484±0.028	136.8±59.9	0.1644±0.030	91.6±59.2	0.4483±0.038	32.5±32.9	0.0544±0.017	132.7±46.4
RandomK	—	0.1035±0.038	267.2±104.2	0.1110±0.042	231.3±111.7	0.3151±0.091	275.9±208.8	0.0403±0.018	183.8±53.8

Table 3: Summary over four synthetic fronts ($d = 3$, $n = 60$, $k = 4$, $\sigma = 0.035$, $T = 3000$, 10 seeds). **Fid.** indicates fidelity to the cited work: • = faithful top-k adaptation (core mechanism unchanged); ○ = our extension (modified core, tie-break, or diversity mechanism not in the original). $^+$ denotes our variant when both versions share a citation. Final Regret values are cumulative α-regret $\pm$ 95% CI.

Additional results for $d = 4$ and $k = 5$

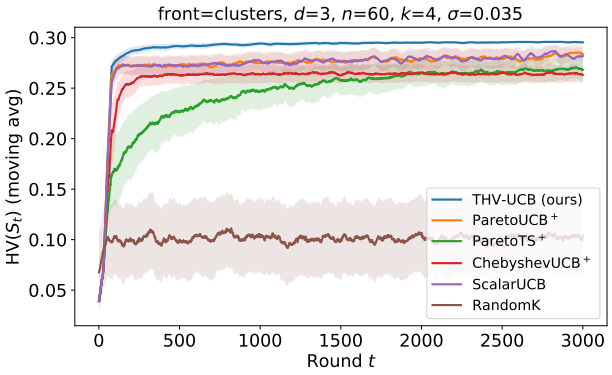
		Clusters		Concave		Convex		Linear	
Algorithm	**Fid.**	$HV_{last100}$	Final Regret	$HV_{last100}$	Final Regret	$HV_{last100}$	Final Regret	$HV_{last100}$	Final Regret
THV-UCB (ours)	—	**0.0854**±0.000	**1.5**±0.0	**0.1051**±0.000	**1.6**±0.0	**0.3709**±0.002	**3.6**±0.0	**0.0677**±0.000	**1.4**±0.1
ParetoUCB (Drugan and Nowe 2013)	•	0.0255±0.009	153.4±44.6	0.0390±0.012	150.9±57.5	0.2369±0.035	150.1±123.6	0.0129±0.007	150.5±32.3
ParetoUCB$^+$ (Drugan and Nowe 2013)	○	0.0791±0.001	1.6±0.1	0.0947±0.002	4.9±3.1	0.3384±0.006	9.4±1.4	0.0403±0.006	44.2±25.9
ParetoUCB-Div (Drugan and Nowe 2013)	○	0.0407±0.003	80.2±20.8	0.0492±0.003	95.4±19.7	0.2564±0.009	19.9±20.6	0.0199±0.007	123.2±35.2
ParetoUCB-Crowd (Drugan and Nowe 2013; Deb et al. 2002)	○	0.0109±0.003	168.5±22.6	0.0157±0.006	215.0±28.5	0.2388±0.019	55.1±41.9	0.0026±0.002	170.3±11.7
ParetoTS (Yahyaa and Manderick 2015)	•	0.0302±0.010	121.3±44.7	0.0468±0.012	112.2±53.0	0.2456±0.033	98.2±93.1	0.0218±0.008	108.0±35.4
ParetoTS$^+$ (Yahyaa and Manderick 2015)	○	0.0749±0.003	14.1±9.6	0.0916±0.003	10.6±8.1	0.3365±0.006	4.5±1.5	0.0390±0.007	48.7±27.6
ChebyshevUCB (Mandow, Martín-Albo, and Perez-de-la Cruza 2023)	•	0.0211±0.015	183.7±57.1	0.0556±0.008	73.2±24.0	0.2349±0.020	76.4±27.9	0.0279±0.005	81.5±20.1
ChebyshevUCB$^+$ (Mandow, Martín-Albo, and Perez-de-la Cruza 2023)	○	0.0800±0.000	**1.5**<0.001	0.0989±0.000	**1.6**<0.001	0.3503±0.004	5.5±1.4	0.0510±0.001	1.9±0.5
HVScalarUCB (Zhang and Golovin 2020)	•	0.0384±0.011	115.8±50.0	0.0539±0.012	103.0±50.7	0.2510±0.029	77.9±68.3	0.0290±0.008	91.1±36.9
HVScalarUCB$^+$ (Zhang 2024)	•	0.0639±0.006	24.3±19.9	0.0852±0.006	5.4±5.6	0.3041±0.020	13.2±15.7	0.0532±0.005	9.7±9.1
ScalarUCB (Auer, Cesa-Bianchi, and Fischer 2002)	○	0.0791±0.000	1.7±0.2	0.0946±0.002	5.8±3.5	0.3379±0.006	7.7±4.1	0.0386±0.007	41.8±25.4
ScalarUCB-RandW (Paria, Kandasamy, and Póczos 2020)	○	0.0353±0.011	100.7±47.3	0.0444±0.014	118.9±58.9	0.2597±0.030	67.9±73.1	0.0206±0.007	116.6±31.9
RandomK	—	0.0168±0.008	186.6±41.9	0.0255±0.012	203.0±58.7	0.1529±0.048	427.7±222.2	0.0098±0.006	164.4±29.6

Table 4: Summary over four synthetic fronts ($d = 4$, $n = 100$, $k = 5$, $\sigma = 0.025$, $T = 5000$, 10 seeds). **Fid.** indicates fidelity to the cited work: • = faithful top-k adaptation (core mechanism unchanged); ○ = our extension (modified core, tie-break, or diversity mechanism not in the original). $^+$ denotes our variant when both versions share a citation. Final Regret values are cumulative α-regret ± 95% CI.

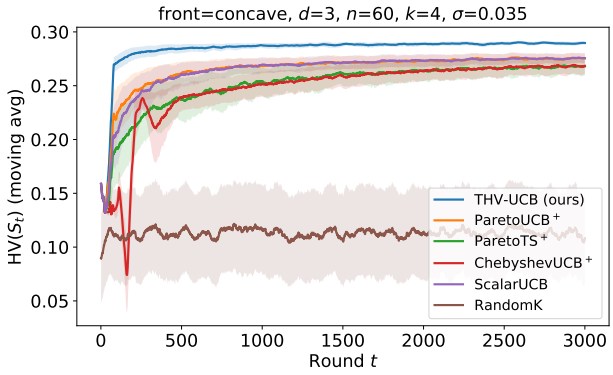
Additional results for $d = 5$ and $k = 6$

		Clusters		Concave		Convex		Linear	
Algorithm	**Fid.**	$HV_{last100}$	Final Regret	$HV_{last100}$	Final Regret	$HV_{last100}$	Final Regret	$HV_{last100}$	Final Regret
THV-UCB (ours)	—	**0.0266**<0.001	**0.7**±0.0	**0.0202**<0.001	**0.6**±0.0	**0.1841**±0.001	**3.7**<0.001	**0.0095**<0.001	**0.3**±0.0
ParetoUCB (Drugan and Nowe 2013)	•	0.0042±0.002	64.7±10.6	0.0032±0.002	48.6±8.0	0.0774±0.018	229.0±86.2	0.0010±0.001	26.1±3.1
ParetoUCB$^+$ (Drugan and Nowe 2013)	○	0.0254<0.001	0.8±0.1	0.0197<0.001	0.6±0.1	0.1789<0.001	3.7<0.001	0.0051±0.001	8.6±4.6
ParetoUCB-Div (Drugan and Nowe 2013)	○	0.0141±0.001	18.8±5.8	0.0094±0.001	22.3±4.7	0.1144±0.012	65.3±29.6	0.0020±0.001	21.4±4.0
ParetoUCB-Crowd (Drugan and Nowe 2013; Deb et al. 2002)	○	0.0025±0.001	64.4±5.3	0.0015±0.001	51.4±4.9	0.0779±0.010	176.9±48.1	0.0015±0.001	24.7±2.3
ParetoTS (Yahyaa and Manderick 2015)	•	0.0057±0.003	56.8±12.3	0.0046±0.002	41.1±9.0	0.0853±0.017	171.8±76.6	0.0016±0.001	23.3±3.9
ParetoTS$^+$ (Yahyaa and Manderick 2015)	○	0.0234±0.001	8.1±4.0	0.0173±0.001	7.3±3.7	0.1716±0.004	7.5±3.9	0.0050±0.001	11.3±5.0
ChebyshevUCB (Mandow, Martín-Albo, and Perez-de-la Cruza 2023)	•	0.0033±0.003	69.9±10.7	0.0092±0.001	19.4±3.2	0.0983±0.008	111.9±18.1	0.0047±0.001	7.9±2.9
ChebyshevUCB$^+$ (Mandow, Martín-Albo, and Perez-de-la Cruza 2023)	○	0.0208<0.001	**0.7**<0.001	0.0150±0.002	5.9±3.2	0.1737±0.003	6.3±1.9	0.0016<0.001	24.0±2.3
HVScalarUCB (Zhang and Golovin 2020)	•	0.0194±0.003	9.5±9.4	0.0123±0.002	13.2±9.1	0.1334±0.017	46.1±47.0	0.0029±0.001	17.1±5.5
HVScalarUCB$^+$ (Zhang 2024)	•	0.0218±0.002	2.6±2.5	0.0151±0.002	4.3±3.9	0.1434±0.011	12.4±12.5	0.0053±0.001	8.1±4.8
ScalarUCB (Auer, Cesa-Bianchi, and Fischer 2002)	○	0.0254<0.001	0.8±0.1	0.0197<0.001	**0.6**±0.1	0.1790<0.001	**3.7**±0.0	0.0049±0.001	9.0±4.6
ScalarUCB-RandW (Paria, Kandasamy, and Póczos 2020)	○	0.0103±0.004	38.9±16.6	0.0077±0.003	31.0±12.3	0.1222±0.020	73.4±57.5	0.0027±0.001	18.8±5.7
RandomK	—	0.0028±0.002	69.9±9.3	0.0026±0.002	51.5±7.4	0.0473±0.019	351.6±93.0	0.0008±0.001	27.0±2.8

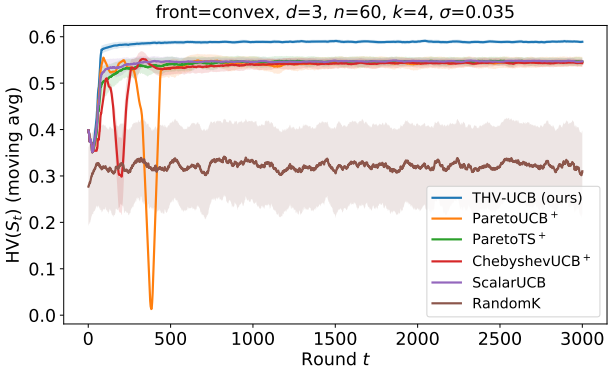
Table 5: Summary over four synthetic fronts ($d = 5$, $n = 150$, $k = 6$, $\sigma = 0.02$, $T = 5000$, 10 seeds). **Fid.** indicates fidelity to the cited work: • = faithful top-k adaptation (core mechanism unchanged); ○ = our extension (modified core, tie-break, or diversity mechanism not in the original). $^+$ denotes our variant when both versions share a citation. Final Regret values are cumulative α-regret ± 95% CI.



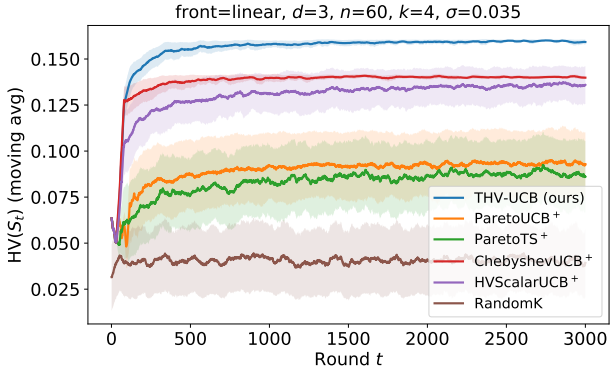
(a) Clusters



(b) Concave

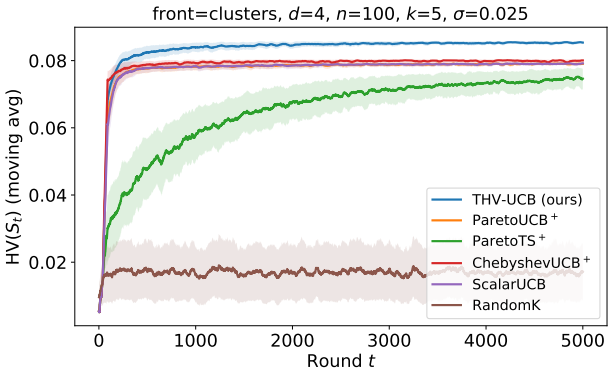


(c) Convex

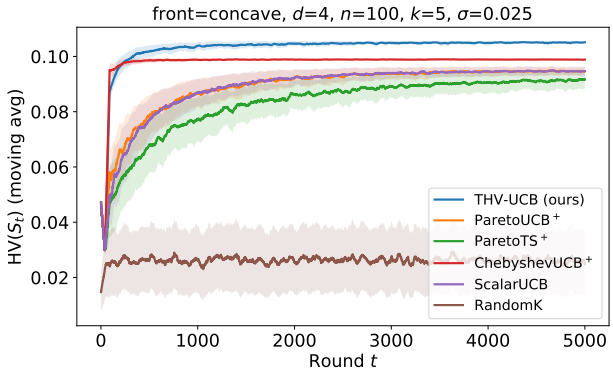


(d) Linear

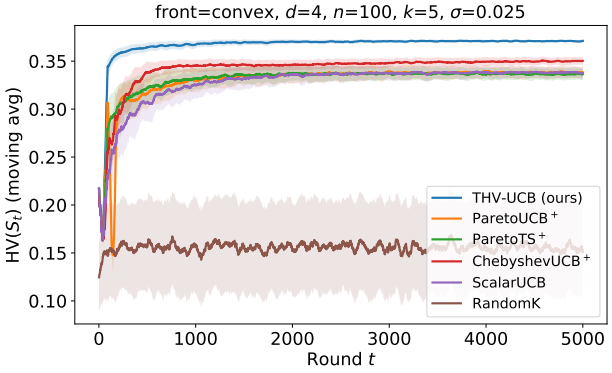
Figure 5: Hypervolume $\mathrm{HV}(S_t)$ trajectories (moving average, $w = 50$) with 95% CIs, over four synthetic Pareto front geometries ($d = 3$, $n = 60$, $k = 4$, $\sigma = 0.035$, $T = 3000$, 10 seeds). Each panel shows the best representative per algorithm family against THV-UCB (ours).



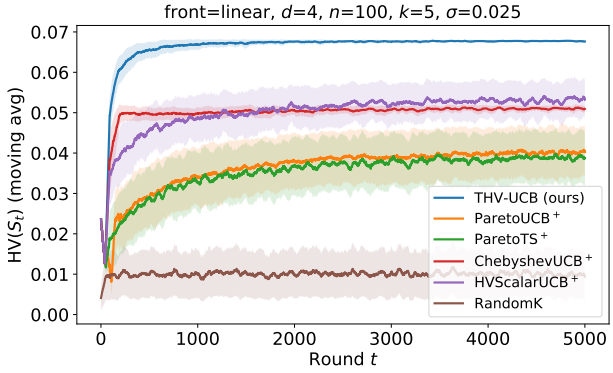
(a) Clusters



(b) Concave

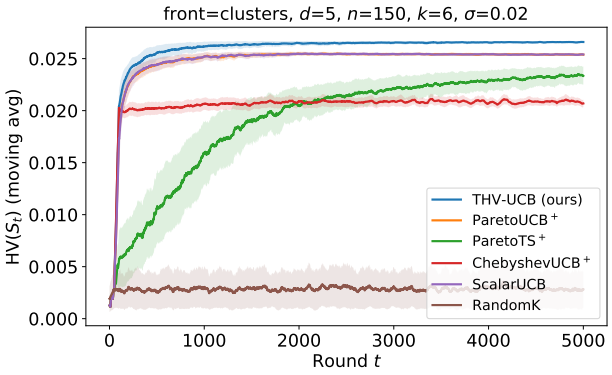


(c) Convex

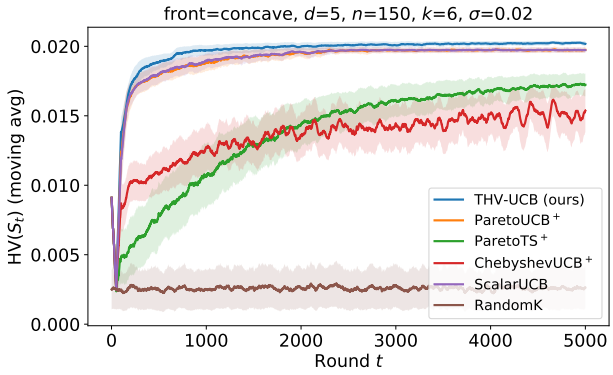


(d) Linear

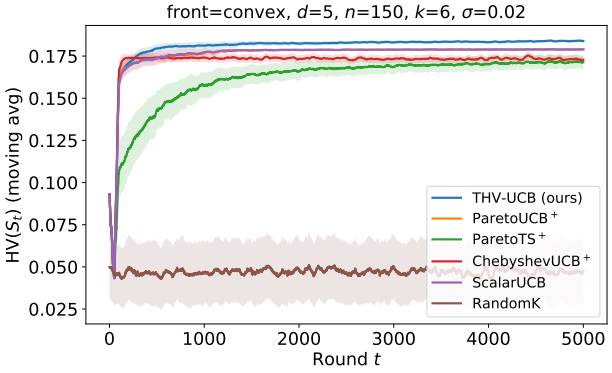
Figure 6: Hypervolume $\mathrm{HV}(S_t)$ trajectories (moving average, $w = 50$) with 95% CIs, over four synthetic Pareto front geometries ($d = 4$, $n = 100$, $k = 5$, $\sigma = 0.025$, $T = 5000$, 10 seeds). Each panel shows the best representative per algorithm family against THV-UCB (ours).



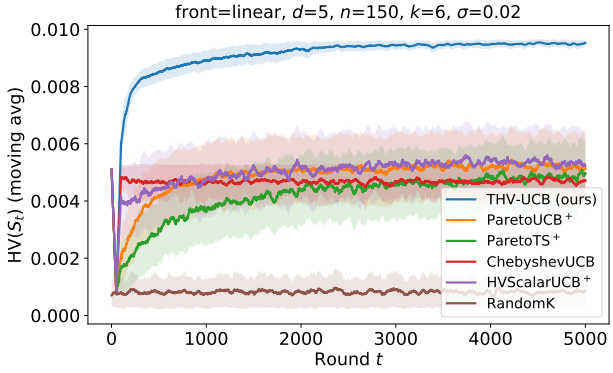
(a) Clusters



(b) Concave



(c) Convex



(d) Linear

Figure 7: Hypervolume $\mathrm{HV}(S_t)$ trajectories (moving average, $w = 50$) with 95% CIs, over four synthetic Pareto front geometries ($d = 5$, $n = 150$, $k = 6$, $\sigma = 0.02$, $T = 5000$, 10 seeds). Each panel shows the best representative per algorithm family against THV-UCB (ours).

Statistical tests

Setting	Baseline	Δ	Bootstrap 95% CI	Cohen's d	Wilcoxon p	Wins
$d = 2$, clusters	ParetoUCB-Div	$+0.0048$	$[+0.0034, +0.0063]$	1.92	< 0.001	$10/10$
$d = 3$, concave	ChebyshevUCB$^+$	$+0.0169$	$[+0.0161, +0.0177]$	12.32	< 0.001	$10/10$
$d = 3$, convex	ParetoTS$^+$	$+0.0430$	$[+0.0418, +0.0442]$	20.45	< 0.001	$10/10$
$d = 4$, clusters	ChebyshevUCB$^+$	$+0.0054$	$[+0.0053, +0.0055]$	31.77	< 0.001	$10/10$
$d = 4$, concave	ChebyshevUCB$^+$	$+0.0062$	$[+0.0061, +0.0063]$	44.48	< 0.001	$10/10$
$d = 4$, linear	HVScalarUCB$^+$	$+0.0145$	$[+0.0141, +0.0148]$	22.84	< 0.001	$10/10$
$d = 5$, clusters	ScalarUCB	$+0.0012$	$[+0.0012, +0.0012]$	29.98	< 0.001	$10/10$
$d = 5$, concave	ScalarUCB	$+0.0005$	$[+0.0005, +0.0005]$	8.09	< 0.001	$10/10$
$d = 5$, convex	ScalarUCB	$+0.0051$	$[+0.0049, +0.0053]$	14.97	< 0.001	$10/10$
$d = 5$, linear	HVScalarUCB$^+$	$+0.0042$	$[+0.0041, +0.0044]$	19.55	< 0.001	$10/10$

Table 6: Paired statistical comparisons between THV-UCB and the second-best baseline on the settings with the tightest margins (10 random seeds, paired by environment seed). Δ denotes the mean difference in $\mathrm{HV}_{\mathrm{last100}}$ (THV-UCB $-$ baseline). Bootstrap 95% CIs are computed over $B = 10{,}000$ resamples. Wilcoxon signed-rank tests are one-sided (H_1: THV-UCB $>$ baseline); *Wins* = number of seeds (out of 10) on which THV-UCB outperforms the baseline.

Notation summary

Symbol	Meaning
n	Number of arms
T	Time horizon (rounds)
d	Number of objectives (reward dimensions)
k	Subset size selected each round
$[n]$	$\{1, \dots, n\}$
$[T]$	$\{1, \dots, T\}$
$S_t \subseteq [n]$	Slate (set) of k arms chosen at round t
$\mathbf{X}_{i,t} \in [0,1]^d$	Reward vector of arm i at round t
$\boldsymbol{\mu}_i = \mathbb{E}[\mathbf{X}_{i,t}]$	Mean reward vector of arm i
$\widehat{\boldsymbol{\mu}}_i(t)$	Empirical mean of arm i up to round t
$N_i(t)$	Number of times arm i was selected up to round t
$\mathcal{F}_t$	Filtration generated by observations up to round t
$\mathbf{r} \in \mathbb{R}^d$	Reference point for hypervolume (dominated by all means)
$\mathrm{HV}(S)$	Dominated hypervolume of set S
$\mathcal{P}$	Pareto front (set of undominated means)
$\preceq, \prec$	(Strict) Pareto dominance relations
η	Confidence parameter; also governs sub-Gaussianity of reward coordinates
$\beta_i(t)$	Confidence radius for arm i at round t
$S^\star \in \arg\max_{\lvert S\rvert=k} \mathrm{HV}(S)$	Best size-k subset in hindsight
$V^\star = \mathrm{HV}(S^\star)$	Benchmark hypervolume value
$G^\star = \{i_1^\star, \dots, i_k^\star\}$	Greedy benchmark on true means
$G_\ell^\star$	Length-ℓ prefix of the greedy benchmark $G^\star$
$\Delta(i \mid S) = \mathrm{HV}(S \cup \{i\}) - \mathrm{HV}(S)$	True marginal hypervolume gain
$\Delta_\ell(i)$	Stage-ℓ gap of arm i relative to $G^\star$
$\Delta_{\min} = \min_{\ell \in [k]} \min_{i \neq i_\ell^\star} \Delta_\ell(i)$	Minimum stage-wise gap
C_d	Coordinate-wise Lipschitz constant of HV ($C_d \leq d$)
$\mathcal{E}$	High-probability event on which all confidence intervals hold
$r_t = V^\star - \mathrm{HV}(S_t)$	Instantaneous regret w.r.t. $V^\star$
$R_T = \sum_{t=1}^T r_t$	Cumulative regret
α	Greedy approximation factor $(1 - 1/e)$
$\bar{r}_t = \alpha V^\star - \mathrm{HV}(S_t)$	Instantaneous α-regret
$\bar{R}_T = \sum_{t=1}^T \bar{r}_t$	Cumulative α-regret
$\lVert \cdot \rVert_\infty$	ℓ_∞ norm

Table 7: Notation summary.